

# SERINA SAKHARE

*Electronics & Hardware Engineer — FPGA / RTL, Analog CMOS, Signal Processing | B.Tech ECE, 2027*

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## PROFILE SUMMARY

ECE undergraduate with **two ISRO (SAC) internships** spanning digital FPGA design and analog CMOS circuit design. Delivered a configurable **BPSK modulator on a Kintex-7 FPGA** for NavIC's Two-Way Messaging Service and a **65 nm two-stage op-amp with 2.72 GHz UGB and 71° phase margin**. Hands-on across the hardware cycle — architecture, RTL, timing closure, ILA debug and bench validation — targeting electronics design, embedded hardware and V&V roles in aerospace, communications and semiconductors.

## TECHNICAL EXPERIENCE

**Summer Intern, Navigation Receiver Division — Space Applications Centre (SAC), ISRO** *May – Jul 2025*

- Designed a **configurable BPSK modulator** on a **Kintex-7 FPGA** (Verilog, Xilinx Vivado) for NavIC's Two-Way Messaging Service (TWMS, planned from NVS-06), integrating **4 custom blocks** — DDS sine-wave generator, FIFO, DAC interface and modulator.
- Generated a BPSK signal at a **12.295 MHz carrier** with **245.88 kHz bandwidth** at a **122.44 kHz data rate**.
- Matched data rates between the processor and modulator using **FIFO buffering**; interfaced the **FMC150 DAC card** with **DDR** transfer.
- Configured **MMCM IP** for on-chip clock synthesis and a **CDCE external clock generator** for reference clocking; closed timing violations with constraints and Vivado Tcl, and debugged post-implementation signals using **ILA**.
- Validated on **oscilloscope and spectrum analyser** — confirmed bit-transition phase reversals and the expected DSB sinc spectrum.

**Winter Intern, Human Spaceflight & Advanced Technology Area — SAC, ISRO** *Dec 2025 – Jan 2026*

- Designed a **two-stage CMOS operational amplifier** (differential pair, current mirrors, gain and buffer stages) in a **65 nm foundry PDK** using Cadence Virtuoso at 3.3 V; laid out the common-source stage to design rules.
- Achieved **35 dB open-loop gain, 2.72 GHz UGB, 71° phase margin and 9.54 dB gain margin** with a common-source second stage, verified through AC, transient and stability analysis.
- Benchmarked **2 second-stage topologies**: common-source (35 dB, high gain) vs. source follower (18 dB, 4.2 MHz UGB, low output impedance), quantifying the gain vs. output-impedance trade-off.

## RELEVANT PROJECTS

**Costas Loop PM Demodulator on FPGA — Independent Project** *Jun – Jul 2026*

- Built a fully digital **PLL-based PM demodulator** in VHDL (Vivado) using **DDS Compiler and FIR Compiler IP** with custom control logic over **AXI4-Stream**; recovered a clean **32 kHz message** from a PM carrier (kp = π).
- Applied **bandpass sampling at fs = 5 MHz** to alias a **69 MHz S-band IF** carrier to a 1 MHz digital carrier; **33-tap symmetric FIR filters** on the I/Q arms rejected the 2fc image.
- Implemented a PI loop filter (damping 0.7) driving a **32-bit phase-accumulator NCO** in fixed point (Q1.15 data, Q1.31 loop filter); RTL output matched the **MATLAB/Simulink** reference model in **both test cases**.

**Queue Management for QoS Enhancement — Independent Project (NS-3)** *Feb – Apr 2026*

- Simulated congestion across network nodes in NS-3, comparing **3 scheduling schemes** — FIFO, Weighted Round Robin and priority queueing — under varying traffic loads; evaluated throughput, latency and packet delivery ratio.

**Smart Healthcare Monitor — Independent Project (IoT / Embedded)** *Sep – Oct 2025*

- Built an **ESP32**-based real-time monitor tracking **4 parameters** (SpO2, heart rate, and temperature and humidity via DHT11), streamed wirelessly to a **Blynk** mobile dashboard.

## EDUCATION & CERTIFICATIONS

**B.Tech, Electronics & Communication Engineering — Manipal Institute of Technology, Manipal** *2023 – 2027*

India Semiconductor Workforce Development Program — IISc | Synopsys | Samsung (**Fellowship Grade II**)

## CORE SKILLS & TOOLS

**Digital / FPGA:** Verilog, VHDL, RTL design & testbenches, Xilinx IP (DDS, FIR, FIFO, MMCM), AXI4-Stream, timing constraints, ILA debug
**Analog / Circuits:** CMOS amplifier design (65 nm PDK), current mirrors, AC / transient / stability analysis, layout to design rules
**Signal Processing & Comms:** BPSK, PM, PLL / Costas loop, DDS / NCO, FIR filtering, bandpass sampling, fixed-point arithmetic
**Hardware & Lab:** Oscilloscope, spectrum analyser, DAC interfacing (FMC150), clock ICs (CDCE), ESP32, sensors
**Tools & Languages:** Xilinx Vivado, Cadence Virtuoso, MATLAB / Simulink, NS-3, Arduino IDE; Python, C, C++, Tcl, Linux (WSL)

## LEADERSHIP & COLLABORATION

- **Treasurer**, Economics & Finance Society of Manipal — managed budgeting and financial planning for a **40+ member** society.
- **Vice President**, Aaina Dramatics Club — led a **40+ member** team at national-level competitions (BITS Goa, IIM Bangalore).

# Design and Implementation of Configurable BPSK Modulator for TWMS in NavIC

SRTD-RTMG-MISA

**Name: Serina Sakhare**
**Designation: Research Intern**
**Institution: Manipal Institute of Technology**

**Guide: Sri. Rakesh Kumar Bijarniya, Mrs Saumi S.**
**Contact e-mail: rakeshkb@sac.isro.gov.in, de_saumi@sac.isro.gov.in**



**21 MAY 2025 to 21 JUL 2025**

## MAJOR OBJECTIVES:

1) Implementation of FIFO for matching data rate between data generator (processor) and modulator.
2) Design and implementation of Sine Wave Generator.
3) Interfacing of Digital to Analog Converter (DAC) based on FMC150 card.
4) Design of a BPSK modulator integrating above modules.

## METHODOLOGY FLOW CHART:

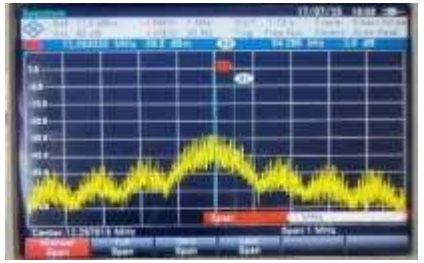
1) Architecture Design
2) Design and implementation of a sine wave generator, FIFO Block, DAC Interface block.
3) Verification and debugging using RTL simulation.
4) Timing constraints and Mapping of IO's
5) Synthesis and Implementation.
6) Debugging of signals post implementation using ILA (Integrated Logic Analyzer).
7) Analysis and verification of the results obtained from hardware using oscilloscope and spectrum analyzer.

## RESULTS/MAJOR FINDINGS:

- ✓ Achieved data rate matching using FIFO
- ✓ Successfully interfaced DAC Card with Kintex-7 FPGA board.
- ✓ Obtained a BPSK modulated signal at carrier frequency of 12.295 MHz with a bandwidth of 245.88 KHz. Data rate and carrier frequency are configurable.
- ✓ Achieved Dual data rate transfer of data between FMC150 and FPGA board.
- ✓ Verified the results in time domain as well as frequency domain using oscilloscope and spectrum analyzer.
- ✓ This work can potentially be used in receiver design for Two Way Messaging Service (TWMS) planned to be offered from NVS-06 onwards.



Time domain capture



Frequency spectrum (data rate: 122.44KHz)

**CONCLUSION:** Project delivered a Binary Phase Shift keying modulator. Double Side Band, Sinc function was successfully obtained on the spectrum analyzer. Corresponding Time domain waveform was also obtained showing phase difference for bit transitions.

# Implementation of PM Demodulator – Costas Loop in FPGA

**Name: Serina Sakhare**
**Designation: Independent Project**
**(B.Tech, Electronics & Communication)**
**Institution: Manipal Institute of Technology, Manipal**

**Project Type:**
**Independent Project**



**01 JUN 2026 to 31 JUL 2026**

## MAJOR OBJECTIVES:

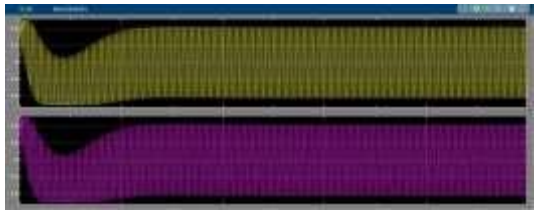
- Design and implement a digital Phase Modulation (PM) demodulator on an FPGA using VHDL and Xilinx Vivado, based on a digital Phase-Locked Loop (PLL) architecture.
- Realise the digital mixers, phase detector, and PI loop filter using custom VHDL and Xilinx IP cores (DDS Compiler, FIR Compiler).
- Use bandpass sampling to translate a 69 MHz S-band IF carrier to an equivalent 1 MHz digital carrier.
- Verify the FPGA (Vivado) implementation against a MATLAB/Simulink reference model.

## METHODOLOGY FLOW CHART:

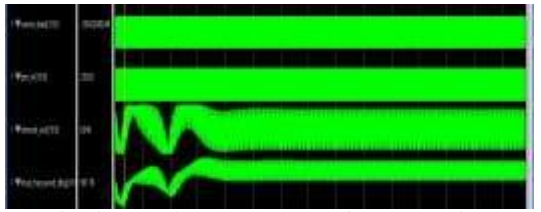
1. System architecture: Antenna (2.2 GHz) → RF down-converter (S-band, 70 MHz) → ADC bandpass sampling (fs = 5 MHz) → PM demodulator.
2. Down-conversion: multiply PM_in by DDS-generated cos/sin to obtain I and Q baseband components.
3. Low-pass filtering: 33-tap symmetric FIR filters on I and Q arms to reject the 2fc image.
4. Phase detection and PI loop filter (coefficients G1, G2, G3; damping factor 0.7) to close the PLL.
5. Loop DDS (streaming NCO, 32-bit phase accumulator) updates every clock cycle to track the phase error.
6. Fixed-point VHDL implementation in Vivado (Q1.15 data, Q1.31 loop-filter output).
7. Verification against MATLAB/Simulink using two test cases: constant input and 32 kHz PM signal.

## RESULTS/MAJOR FINDINGS:

- ✓ Achieved PLL lock-on with a constant input; loop filter output converged smoothly to steady state.
- ✓ Bandpass sampling aliased the 69 MHz RF carrier to a 1 MHz digital carrier at fs = 5 MHz with spectral reversal, as designed.
- ✓ Recovered a clean 32 kHz message signal from a PM carrier (kp = π) on the FPGA after PLL lock.
- ✓ Vivado simulation waveforms closely matched the MATLAB/Simulink reference model for both test cases.
- ✓ Implemented using Xilinx DDS Compiler and FIR Compiler IP cores with custom control logic over AXI4-Stream.
- ✓ This work demonstrates a flexible, fully digital PM demodulation architecture for FPGA-based communication receivers.



*MATLAB/Simulink scope (32 kHz PM signal)*



*Vivado waveform — recovered message (PM signal)*

**CONCLUSION:** Project delivered a fully digital PM demodulator on FPGA using a PLL architecture (DDS Compiler + FIR Compiler + custom VHDL). The design recovered a 32 kHz message signal from a phase-modulated carrier, and Vivado simulation results closely matched the MATLAB/Simulink reference model for both the constant-input and PM-signal test cases.

# Design and Analysis of Two-Stage Operational Amplifier using 65nm CMOS Process

AVDD/QUTG/HSTA — SAC-ISRO

**Name: Serina Sakhare (Reg. No. 230907084)**
**Designation: B.Tech Intern (Electronics & Communication)**
**Institution: Manipal Institute of Technology, Manipal**

**Guide: Dr. Hari Shanker Gupta (Guide), Mr. Sanjay Kasodniya (Co-Guide)**



**02 DEC 2025 to 05 JAN 2026**

## MAJOR OBJECTIVES:

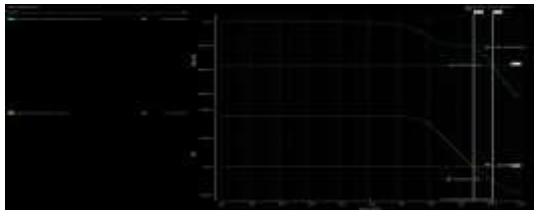
- Design and schematic-level analysis of MOSFET-based building blocks — common-source and common-drain (source-follower) stages — in 65 nm CMOS.
- Design of a two-stage differential amplifier combining a differential input pair with a common-source or common-drain second stage.
- Characterisation of AC response, gain margin, phase margin, unity-gain bandwidth (UGB), and transient response using Cadence Virtuoso.
- Comparison of common-source vs. common-drain second-stage configurations for gain / output-impedance trade-offs.

## METHODOLOGY FLOW CHART:

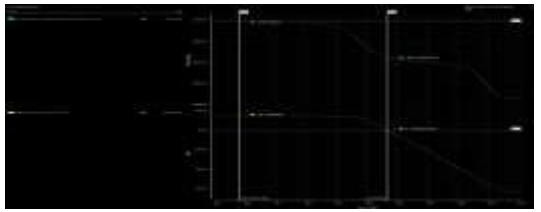
1. Design and simulate common-source and common-drain (source-follower) stages individually in 65 nm NMOS.
2. Layout the common-source stage using MOS_N33 devices, adhering to design rules.
3. Set up the differential-pair test bench (Vin1/Vin2 = 0.01 V AC, 1 V DC, 100 kHz, Vdd = 3.3 V, Vbias = 0.75 V).
4. Cascade the differential pair with a common-drain second stage; simulate AC and transient response.
5. Cascade the differential pair with a common-source second stage; simulate AC and transient response.
6. Extract gain, gain margin, phase margin and UGB from the AC sweeps for both configurations.
7. Compare results between configurations and validate against predicted values.

## RESULTS/MAJOR FINDINGS:

- ✓ Common-source stage AC gain rolled off cleanly at high frequency, with amplification confirmed by transient simulation.
- ✓ Source-follower (common-drain) stage confirmed low output impedance, with gain just under unity as expected.
- ✓ Differential amp with common-drain 2nd stage: UGB 4.2 MHz, Gain 18 dB, Gain Margin 18.45 dB, Phase Margin −83°.
- ✓ Differential amp with common-source 2nd stage: UGB 2.72 GHz, Gain 35 dB, Gain Margin 9.54 dB, Phase Margin 71°.
- ✓ Both configurations showed stable, clean sinusoidal transient response matching predicted behaviour.
- ✓ Results confirm the differential amplifier as a strong candidate module for integration into larger amplification front-ends.



*AC gain/phase — common-source 2nd stage*



*AC gain/phase — common-drain 2nd stage*

**CONCLUSION:** Project delivered a two-stage differential amplifier in 65 nm CMOS, characterising common-source and common-drain second-stage options. Simulation in Cadence Virtuoso confirmed stable gain and phase margins in line with predicted values — the common-source stage offering high gain, the common-drain stage offering low output impedance — together demonstrating suitability for a range of amplification front-ends.